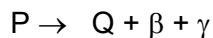


- 7 (a) State what is meant by *half-life*.

.....
 [2]

- (b) A stationary radioactive isotope P decays by emitting a β -particle and γ -radiation. The daughter nucleus produced during this decay is Q. An incomplete equation to represent this decay is



The variation with time t of the number N of undecayed nuclei of radioactive sample of isotope P is shown in Fig. 7.1.

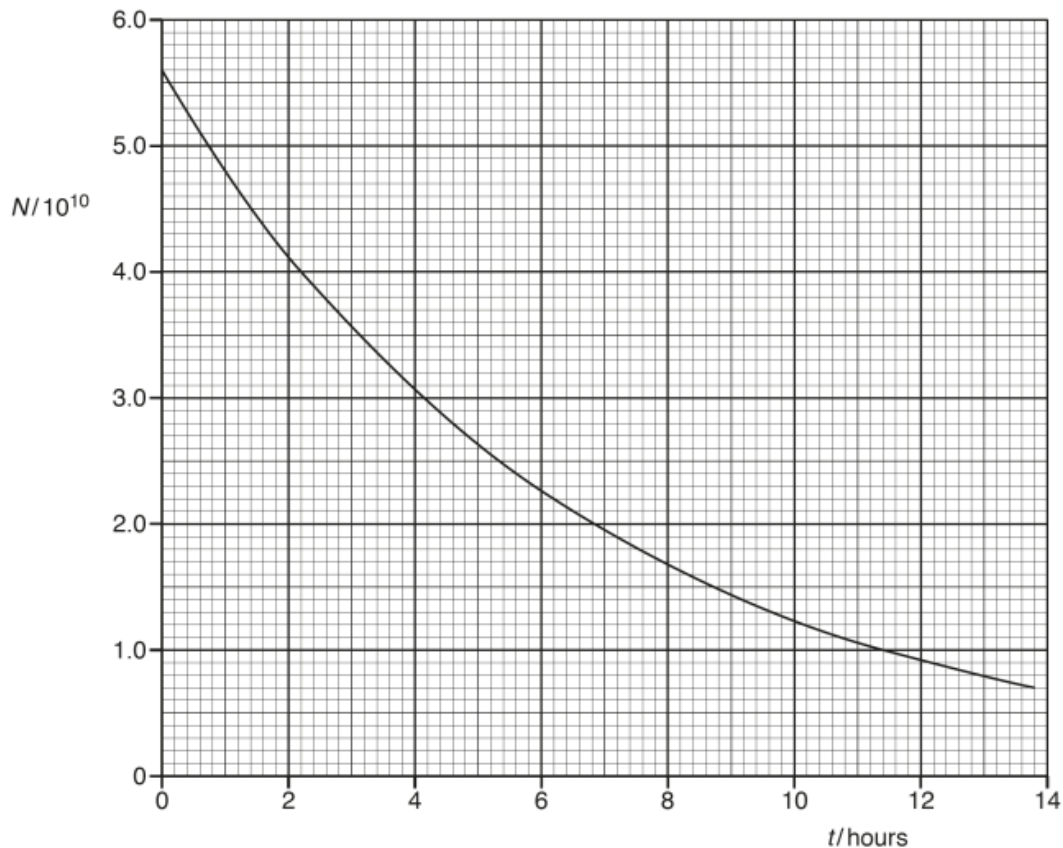


Fig 7.1

- (i) Use Fig. 7.1 to estimate the half-life of isotope P.

half-life =hours [2]

- (ii) Initially, there are no Q nucleus in the sample.
 After a period of time t , the ratio $\frac{\text{number of Q nuclei}}{\text{number of P nuclei}}$ equals 6.
 Calculate t .

$t = \dots\dots\dots$ hours [2]

- (iii) State the significance of the gradient of the graph.

$\dots\dots\dots$
 $\dots\dots\dots$ [1]

- (iv) Determine the activity of isotope P at $t = 4.0$ hours.

activity = $\dots\dots\dots$ Bq [3]

- (v) The daughter nucleus Q is stable. On Fig. 7.1, sketch a graph to show the variation with time t of the number of daughter nuclei Q in the sample. [2]

- (vi) State and explain why the sum of the kinetic energy of the β -particle and the energy of the γ -radiation is less than the total energy released during the decay of isotope P.

$\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$ [3]

- (c) Isotope P is produced in a laboratory by bombarding a stationary nucleus S with an α -particle. It results in the following nuclear reaction.

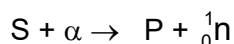


Fig. 7.2 shows the position of isotope P on a diagram in which nucleon number is plotted against proton number. Each small square represents a unit increase in the nucleon number and proton number in the direction of the axes.

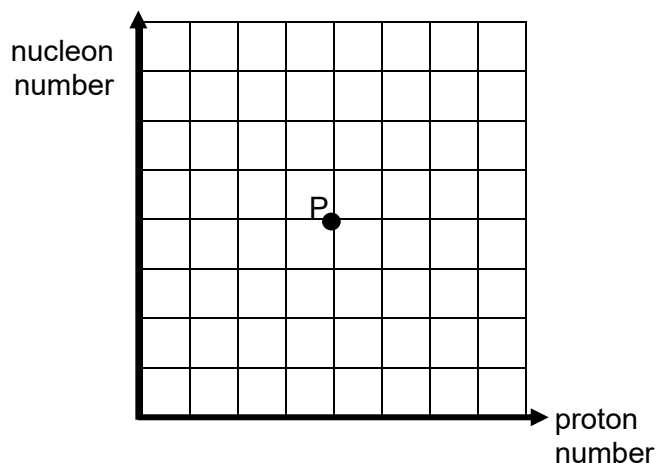


Fig. 7.2

- (i) On Fig. 7.2, mark with the symbol S the position of the nuclide S. [1]
- (ii) With reference to (b), mark on Fig. 7.2 with the symbol Q the position of the daughter nuclide Q due to the decay of isotope P. [1]
- (d) Fig. 7.3 shows an α -particle approaching a stationary gold nucleus head-on. The distance of closest approach of the α -particle to the nucleus S is d .

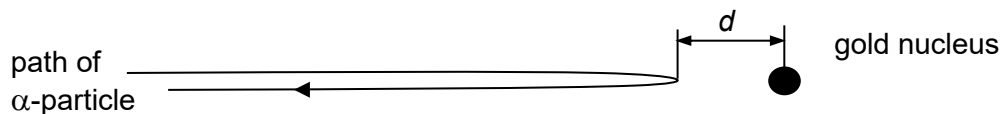


Fig. 7.3

At its distance of closest approach d , explain whether it is possible for the gold nucleus and the α -particle to be at rest simultaneously.

.....

.....

.....

..... [3]